

40 — SAMPLE COMPLEXITY / PAC BOUND CALCULATOR

Aim

To calculate the number of training examples required for PAC learning using the finite hypothesis-space sample complexity bound.

Software Required

- Python 3.x
- Jupyter Notebook / Google Colab
- Python Standard Library (math)

Algorithm + Procedure

1. Define the size of the hypothesis space.
2. Define the error parameter ϵ .
3. Define the confidence parameter δ .
4. Apply the formula:

$$m \geq (1/\epsilon)[\ln|H| + \ln(1/\delta)]$$

5. Calculate the required sample size.
6. Round the value upward.
7. Repeat for different hypothesis-space sizes.
8. Display the required number of training examples.

Code

```
import math

def sample_complexity(
    H_size,
    epsilon,
    delta
):
    m = (
        1 / epsilon
    ) * (
        math.log(H_size)
```

```

        + math.log(1 / delta)
    )
    return math.ceil(m)
for H_size in [10, 100, 1000]:
    m = sample_complexity(
        H_size,
        epsilon=0.1,
        delta=0.05
    )
    print(
        f"|H|={H_size}: "
        f"m >= {m} examples needed"
    )

```

Output

```

|H|=10: m >= 53 examples needed
|H|=100: m >= 77 examples needed
|H|=1000: m >= 100 examples needed

```

Result

The PAC sample complexity bound was successfully calculated for different hypothesis-space sizes. The results demonstrate the logarithmic dependence of sample complexity on the size of the hypothesis space.